

# Iterative Progressive Retrieval Attention: Bounded Associative Closure over Latent Gist Memory

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## Abstract

One-shot semantic retrieval finds memory nearest to the current query. It can miss a second piece of evidence that is weakly related to the query but strongly related to the first piece. We study iterative Progressive Retrieval Attention (PRA), which treats cached routing gists as an implicit graph: the query retrieves a bounded frontier, retrieved gists query the same layer-local index, and only the final closure is mapped to full native key/value (K/V) state. The transformer is not rerun between retrieval rounds.

A parameter-free implementation enforces depth, branch, beam, and unique-chunk budgets; prevents cycles; retains alternate discovery edges; and emits a versioned retrieval graph before K/V materialization. At matched final chunk budgets, controlled two- and three-hop latent graphs show the intended separation: iterative routing completes every chain over 320 paired trials, whereas one-shot Top- $B$  retrieves only the first node. On five independently trained semantic routers and held-out Qwen3-0.6B features, the result is mixed. At 10% selected chunks, HotpotQA evidence-group completion rises from 0.138 to 0.150/0.163/0.175 for depths 2/3/4, but any-evidence recall falls from 0.700 to 0.663/0.663/0.613. QASPER, a mostly direct-evidence control, favors one-shot routing. Search comparisons increase by  $4.4\times$  on HotpotQA and  $9.8\times$  on QASPER at depth 2. Root anchoring reduces drift, and no tested parameter-free state or path update provides a consistent advantage. The findings establish the mechanism and its systems contract, but do not establish that a router trained only for query–evidence relevance learns a useful evidence–evidence graph. Associative-edge supervision is the next test; learned recursion is not yet justified.

## 1 Introduction

Suppose a question is directly related to memory chunk  $A$ . Chunk  $A$ , in turn, identifies an entity or relation needed to find chunk  $B$ . A one-shot router computes similarities from the question to all chunks. It can retrieve  $A$  while missing  $B$  because the question does not yet contain the intermediate relation. Increasing top- $k$  may find  $B$ , but it spends the same extra budget on unrelated chunks. The problem is not transformer reasoning after evidence arrives. It is evidence discovery before full memory becomes active.

PRA already separates compact semantic routing state from full layer-native K/V payloads [8, 7, 9]. This separation suggests a narrow extension. After retrieving  $A$ , use  $A$ ’s compact gist as another query. Continue for a bounded number of rounds, preserve the discovery graph, and materialize exact K/V only once. The resulting path  $q \rightarrow A \rightarrow B$  is explicit. It does not require another language-model pass, an intermediate text answer, or serialization through an external tool API.

This paper asks whether pretrained hidden-state gists already possess enough transitive semantic geometry for that extension to help. The question has a strict baseline: iterative closure must beat one-shot Top- $B$  at the same final unique-chunk budget  $B$ . Otherwise, retrieval depth adds cost without adding a useful primitive.

The contributions are fourfold. First, we formulate iterative PRA as query-conditioned bounded associative closure over an implicit latent graph. Second, we implement Level-1 gist-to-gist and Level-2 aggregate-frontier traversal with deterministic budgets, cycle handling, root anchoring, transparent path scores, and a versioned graph artifact. Third, we evaluate five semantic-router seeds on held-out HotpotQA and QASPER features, together with exact two- and three-hop synthetic controls. Fourth, we report a result that narrows the next experiment: traversal works when associative edges are present, but a router trained for query-evidence ranking does not reliably induce those edges on natural data.

## 2 From Nearest Neighbors to Associative Closure

### 2.1 Why one-shot retrieval can stop too early

Let  $G = \{g_1, \dots, g_N\}$  be compact routing gists at one transformer layer and let  $q$  be the current routing query. One-shot PRA returns

$$N_B(q) = \text{TopB}_{g \in G} s(q, g), \quad (1)$$

where  $s$  is cosine similarity in the learned semantic space. This operation is appropriate when every useful chunk is independently similar to  $q$ . It is incomplete when

$$s(q, g_A) \gg 0, \quad s(q, g_B) \approx 0, \quad s(g_A, g_B) \gg 0. \quad (2)$$

Here  $A$  is an entry point and  $B$  is relationally accessible only after  $A$  is known.

We interpret each chunk as a node in an implicit graph. High gist similarity supplies candidate directed edges, generated lazily when a node becomes part of the frontier. No  $N \times N$  adjacency matrix is built. Iterative PRA therefore computes a bounded semantic neighborhood, not a proof and not complete reasoning.

### 2.2 The complete data path

Figure 1 separates traversal from payload activation. This boundary is the main systems invariant: compact gists may participate in several search rounds, while full K/V cannot enter attention until the closure stops.

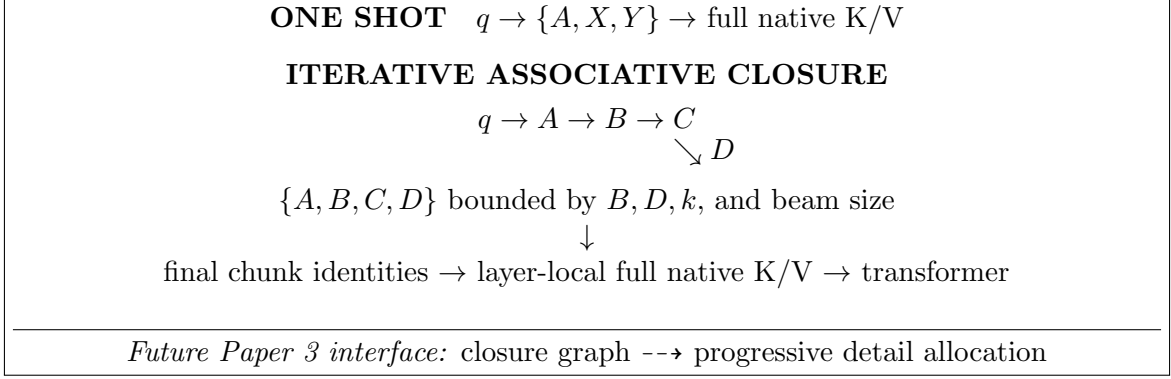


Figure 1: Iterative PRA increases search depth without transformer reruns. Gists are routing state; native K/V is the payload. The dashed handoff preserves graph signals for later work on partial materialization.

### 3 Method

#### 3.1 Bounded frontier expansion

The root query initializes frontier  $F_0 = \{q\}$  and the visited set is empty. At round  $t$ , each frontier vector proposes its  $k$  highest-scoring unseen chunks. Duplicate proposals are merged, the best path to each candidate is retained, and at most the remaining beam and unique-node budgets are accepted. Accepted gists form  $F_t$ . Traversal stops at fixed depth  $D$ , final budget  $B$ , no-new-node convergence, or an optional confidence threshold.

To limit semantic drift, candidate  $c$  is scored against both the root and its frontier parent:

$$e_t(c) = \alpha s(q, g_c) + (1 - \alpha) s(f_{t-1}, g_c), \quad 0 \leq \alpha \leq 1. \quad (3)$$

At  $\alpha = 0$ , traversal follows local gist edges. At  $\alpha = 1$ , every round reproduces root ranking after visited-node removal. Intermediate values preserve the original information need while allowing a retrieved node to redirect search.

The primary path score multiplies affinities  $a_i = (e_i + 1)/2$  along a path:

$$P(c_t) = \prod_{i=1}^t a_i. \quad (4)$$

We also evaluate log-sum, final-edge, minimum-edge, mean-edge, and direct-root scores. These choices are deliberately transparent. A learned controller would confound the first question: whether useful graph structure is already present.

Listing 1: Level-1 bounded associative closure

```

frontier = {root}; selected = {}; graph = empty
for hop in 1..D:
    proposals = union topk_unseen(parent, G, k) for parent in frontier
    merge duplicate targets; retain parent edges and best path
    accepted = top(proposals, min(beam, B - len(selected)))
    add accepted nodes and edges to graph
    frontier = winning_gists(accepted)
    stop if frontier is empty or len(selected) == B
materialize_full_native_kv(selected) # only after closure

```

### 3.2 Level-2 frontier updates

Direct Level-1 traversal uses each winning gist as the next query. We compare three parameter-free updates. A residual update keeps parent state,

$$r_{t+1} = \text{normalize}(r_t + \beta g_t), \quad (5)$$

while mean and path-weighted mean collapse accepted branches to one aggregate vector and add the root. These updates test whether mild state accumulation reduces drift. They do not rerun a transformer and introduce no learned parameters.

### 3.3 Budgets, cycles, and layer locality

Four controls bound expansion: depth  $D$ , neighbors per frontier node  $k$ , per-round beam size, and maximum unique chunks  $B$ . A visited set prevents  $A \rightarrow B \rightarrow A$  from consuming budget twice. Alternate parents remain graph edges even when one path wins pruning. Every index is layer-local. A query, gist, and native payload from different layers are never treated as interchangeable merely because their dimensions match.

If  $C$  is the number of chunks and  $|F_t|$  the frontier size, Level-1 scoring performs  $C \sum_t |F_t|$  gist comparisons. Full K/V transfer remains proportional to the final selected payload, exactly as in matched one-shot PRA. The method trades more compact-index work for a chance to find a better closure without increasing active K/V.

## 4 Experimental Design

### 4.1 Frozen semantic geometry

We reuse Paper 2’s frozen Qwen3-0.6B feature artifact and its attention-input hidden-state representation [5, 9]. The held-out split contains 16 HotpotQA and 16 QASPER examples, 32-token chunks, exact evidence-overlap masks, and 1,024-dimensional root and memory features [11, 2]. Five asymmetric linear projections, each with 262,144 parameters and seeds 11, 23, 37, 53, and 71, map query and memory features to 128 dimensions. They were trained on the disjoint Paper 2 train split with exhaustive negatives. No router is trained or selected on iterative results.

The primary setting uses direct frontier gists, multiplicative path score,  $\alpha = 0.25$ , and  $D \in \{2, 3, 4\}$ . Final budgets are 5, 10, 20, and 30% of each example’s candidate chunks. One-shot always receives the identical final  $B$ . We separately sweep  $\alpha \in \{0, .25, .5, .75, 1\}$ , all Level-2 updates, and all six path scores at  $D = 2$ , 10%.

### 4.2 What the evidence metrics mean

Any-evidence recall asks whether one annotated chunk is selected. Exact all-evidence recall asks whether every evidence-overlapping 32-token chunk is selected. Because one supporting sentence can cross a chunk boundary, we additionally group contiguous positive chunks and define chain completion as selecting at least one chunk from every group. The artifact reports budget feasibility for both definitions. Hotpot groups are ordered by source position; they are not claimed to encode causal hop order. QASPER usually has one contiguous evidence group and is therefore a direct-retrieval control.

The synthetic controls provide exact causal chains. Their root is similar to node  $A$ ; each next node is similar to its predecessor but is outranked from the root by distractors. At matched  $B = 2$

or 3, one-shot can afford the entry node but not the full chain. We run 64 examples per seed for each chain length, yielding 320 paired trials per task.

### 4.3 Downstream execution validation

A separate two-example CUDA smoke loads the pinned Qwen model, indexes one HotpotQA and one QASPER source, and compares no memory, one-shot, and  $D = 2$  closure. One-shot and closure both request three chunks and materialize 96 native K/V tokens. This run validates execution and budget invariants; two examples cannot estimate answer quality.

## 5 Results

### 5.1 Exact latent chains validate the mechanism

Table 1 gives the cleanest mechanism test. Both methods find some evidence, but one-shot retrieves only the entry node. Iterative traversal follows the designed edges and completes every chain. The gain costs one complete index scan per hop.

Table 1: Matched-budget synthetic results over five seeds and 64 examples per seed.

| Task  | Method            | Any   | Exact chain | Coverage | Comparisons |
|-------|-------------------|-------|-------------|----------|-------------|
| 2-hop | One-shot Top-2    | 1.000 | 0.000       | 0.500    | 10          |
| 2-hop | Iterative $D = 2$ | 1.000 | 1.000       | 1.000    | 20          |
| 3-hop | One-shot Top-3    | 1.000 | 0.000       | 0.333    | 11          |
| 3-hop | Iterative $D = 3$ | 1.000 | 1.000       | 1.000    | 33          |

This result establishes that the implementation can recover indirect nodes without transformer updates. It does not show that a pretrained semantic router supplies the required edges.

### 5.2 HotpotQA shows a small chain signal and a larger drift cost

Figure 2 and Table 2 compare natural-data routing. At 10%, one-shot any-evidence recall is  $0.700 \pm 0.065$  (five-seed 95% interval), while  $D = 2$  obtains  $0.663 \pm 0.118$ . Yet group-level chain completion rises from 0.138 to 0.150, and reaches 0.175 at  $D = 4$ . The two metrics describe different behavior: closure sometimes reaches another evidence group, but also leaves an easily retrieved entry neighborhood.

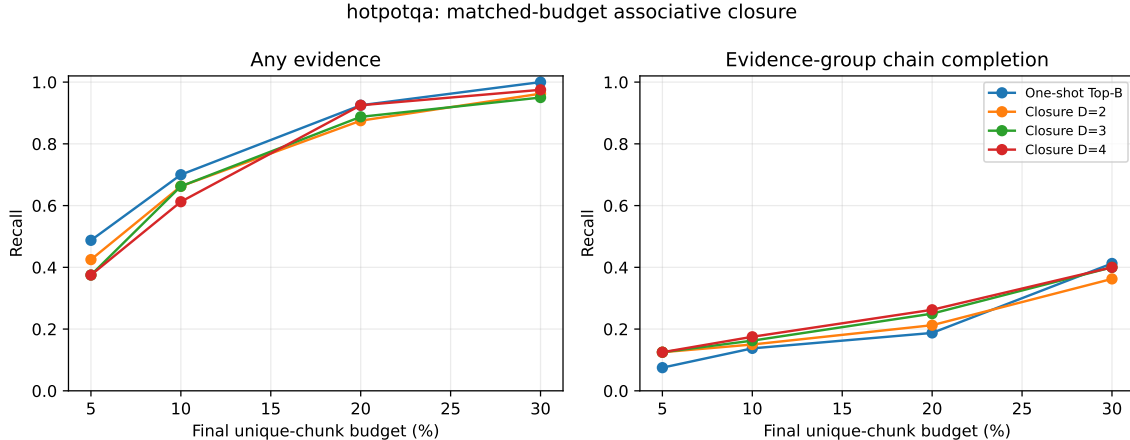


Figure 2: Five-seed HotpotQA matched-budget curves. Deeper closure modestly improves chain completion at some sparse budgets but does not improve the broader any-evidence curve.

Table 2: Primary 10% matched-budget results. Means combine 16 examples and five independently trained routers. “All” requires every positive chunk; “Chain” requires every evidence group.

| Dataset  | Method          | Any  | Chain | All  | Coverage | Comparisons |
|----------|-----------------|------|-------|------|----------|-------------|
| HotpotQA | One-shot        | .700 | .138  | .013 | .207     | 50.1        |
| HotpotQA | Closure $D = 2$ | .663 | .150  | .000 | .202     | 219.6       |
| HotpotQA | Closure $D = 3$ | .663 | .163  | .000 | .205     | 273.0       |
| HotpotQA | Closure $D = 4$ | .613 | .175  | .000 | .193     | 303.2       |
| QASPER   | One-shot        | .963 | .963  | .213 | .477     | 142.6       |
| QASPER   | Closure $D = 2$ | .875 | .875  | .200 | .466     | 1395.9      |
| QASPER   | Closure $D = 3$ | .788 | .788  | .175 | .425     | 1822.1      |
| QASPER   | Closure $D = 4$ | .825 | .825  | .150 | .426     | 2144.2      |

At 20%, Hotpot chain completion rises from 0.188 one-shot to 0.212, 0.250, and 0.263 at depths 2–4. At 30%, one-shot returns to the lead (0.412 versus 0.362–0.400). The effect is therefore small, budget dependent, and not accompanied by improved exact all-chunk recall. Five seeds are insufficient to claim a reliable improvement; the direction also changes with budget.

### 5.3 QASPER behaves as a direct-retrieval control

QASPER’s evidence commonly forms one contiguous group, so chain completion equals any-evidence recall in this protocol. Figure 3 shows that one-shot dominates throughout the sparse region. At 10%, its any/chain recall is  $0.963 \pm 0.069$  versus  $0.875 \pm 0.095$  for  $D = 2$ . Closure adds no useful intermediate relation when the root already addresses the target.

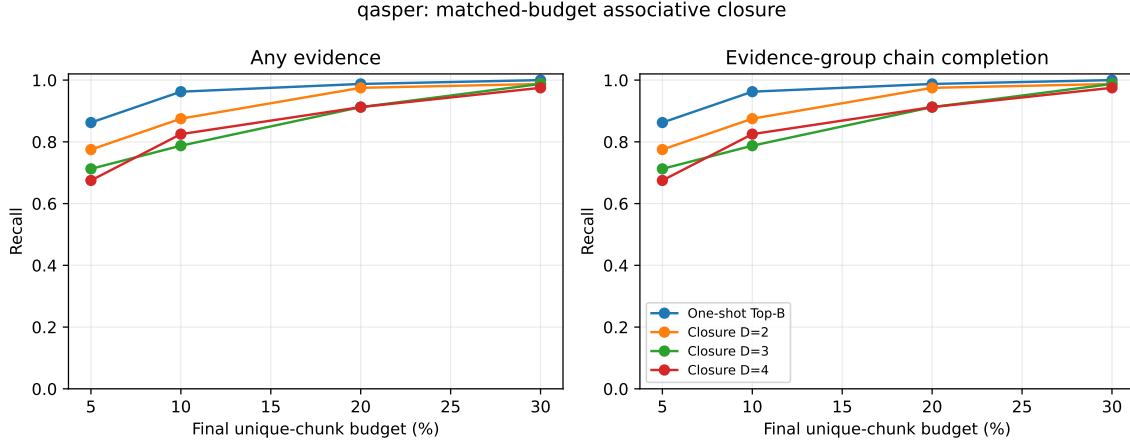


Figure 3: QASPER matched-budget curves. The mostly single-group evidence protocol favors direct one-shot ranking, as expected if recursion is unnecessary.

#### 5.4 Root anchoring exposes semantic drift

At 10% Hotpot, pure local traversal ( $\alpha = 0$ ) yields any recall 0.600. Increasing root weight to 0.50, 0.75, and 1.0 gives 0.700, 0.738, and 0.700. QASPER improves monotonically toward the root-only condition: 0.863, 0.875, 0.900, 0.913, and 0.963 from  $\alpha = 0$  to 1. The best QASPER “iterative” setting is therefore the one that does not use frontier semantics.

Residual update is the only Level-2 variant to preserve exact all-chunk recall on Hotpot (0.013), but its any recall 0.713 does not consistently beat one-shot. Mean and weighted mean reach 0.738 any recall while exact all remains zero. Path-score variants range from 0.650 to 0.738 Hotpot any recall and do not improve chain completion consistently. No parameter-free update dominates.

#### 5.5 Relational gap does not identify real gains

The motivating diagnostic is

$$\Delta_{rel} = \max_{A \in E \setminus \{B\}} s(g_A, g_B) - s(q, g_B). \quad (6)$$

Large  $\Delta_{rel}$  should favor closure if evidence-to-evidence similarity is a useful edge. Figure 4 does not show that pattern. In Hotpot’s highest gap quartile, the paired  $D = 2$  any-recall effect is  $-0.100$  and coverage effect is  $+0.017$ . In QASPER they are  $-0.176$  and  $-0.019$ . A high pairwise cosine score is not enough to identify the correct relational transition.

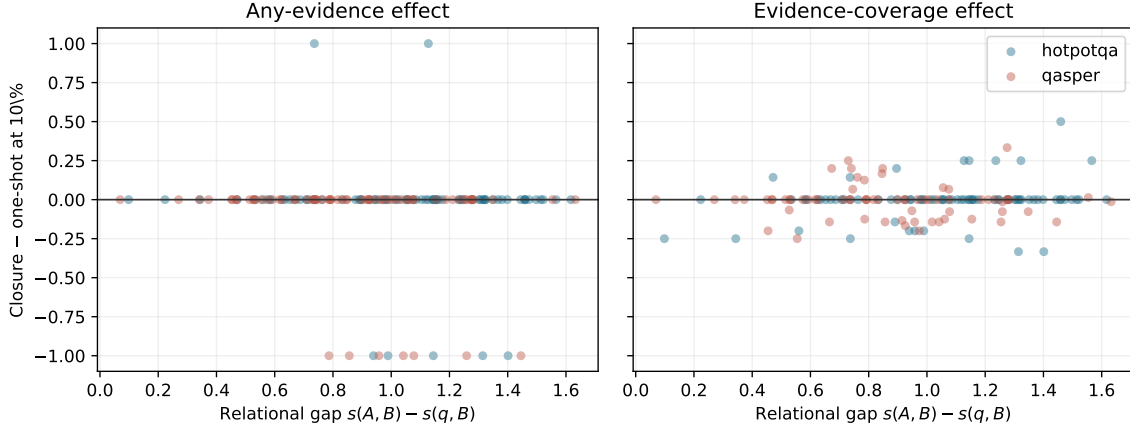


Figure 4: Paired  $D = 2$  minus one-shot effects at 10%. Relational gap does not predict a stable closure advantage in the inherited semantic space.

## 5.6 Search cost and full-K/V execution

Figure 5 shows the compact-index cost. At 10%,  $D = 2$  uses 219.6 comparisons per Hotpot example versus 50.1 one-shot ( $4.4\times$ ), and 1395.9 versus 142.6 on QASPER ( $9.8\times$ ). The higher QASPER multiplier follows from more candidates and larger frontiers. These are exact comparison counts; wall time depends strongly on batching and hardware.

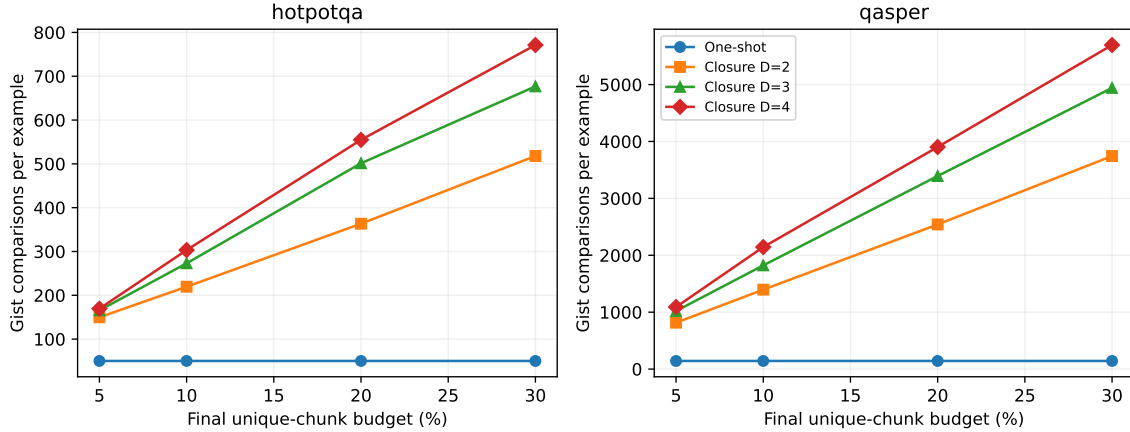


Figure 5: Compact gist comparisons. Iteration leaves final K/V budgets fixed but increases index work with frontier size and depth.

The CUDA integration smoke materializes exactly three chunks and 96 K/V tokens for both methods. Iterative routing takes 0.103/0.077 seconds on the Hotpot/QASPER examples, versus 0.107/0.073 for one-shot in this trace-oriented implementation. End-to-end times are 1.291/1.291 seconds for closure and 1.304/1.347 seconds for one-shot. The model records zero native-limit violations and a maximum 64-token native operation. These two cases validate the execution contract only.



## 6 Discussion

### 6.1 What the experiments establish

The synthetic result and the integration tests establish three mechanical facts. First, a gist can serve as a new query and recover a node that the root cannot afford at matched  $B$ . Second, deduplication, beam pruning, and hard budgets prevent exponential materialization. Third, the closure can remain entirely in compact routing state until final identities map to full native K/V.

The natural-data result establishes a different boundary. Paper 2’s router learned an asymmetric query–memory metric. During iterative traversal, a memory-projected gist is reused as though it were a valid query-projected representation. Shared dimensionality does not imply shared semantics. Even without that asymmetry, evidence supervision optimized root-to-evidence ranking, not evidence-to-next-evidence edges. The mild Hotpot chain signal suggests some useful local structure, but the any-recall loss, unstable budget trend, and relational-gap result show that it is not yet a dependable graph.

The next experiment should therefore train a dual-use or explicitly compositional routing space. Positive edges should connect an entry evidence chunk to the next required evidence chunk, while hard negatives should include semantically related but relation-wrong chunks. Evaluation must remain matched-budget and include a shuffled-edge control. A learned recurrent controller is premature until that simpler geometry test succeeds.

### 6.2 Relation to retrieval, agents, and recursive models

Memory Networks repeatedly read differentiable memory to support multi-hop tasks, while RAG, RETRO, and interleaved retrieval systems bring external evidence into generation [4, 1, 10]. Agentic systems can reformulate a text query after each model pass. Iterative PRA is narrower: it traverses already computed latent addresses inside one memory subsystem. It avoids repeated text serialization and full-model round trips, but it cannot replace an agent that must plan tools, verify evidence, or change the retrieval language.

Tiny Recursive Models repeatedly transform internal latent and answer state [3]. Controlled autoregressive work finds no reliable gain from the full autoregressive TRM architecture over simpler refinement baselines [6]. Iterative PRA does not apply a tiny neural network repeatedly. Its recurrence is *query*  $\rightarrow$  *memory node*  $\rightarrow$  *memory node*. This distinction motivates the parameter-free Level-1 test: retrieval recursion should earn its complexity before neural state recursion is introduced.

### 6.3 Paper 3 handoff

Papers 1–2 conservatively compress search but materialize every selected chunk in full. Iterative closure produces more structure before transfer: hop, parent edges, direct relevance, path score, confidence, branch entropy, and duplicate/overlap statistics. The versioned graph keeps these signals even when the final payload remains unchanged. Paper 3 can therefore ask a separate question: given a discovered closure, how much detail should each node receive? This paper does not test partial materialization.

## 7 Limitations

The held-out natural-data sample has 16 examples per dataset, repeated over five router seeds. Seeds quantify projection sensitivity but do not create 80 independent questions. Hotpot evidence

groups are source-contiguous spans, not annotated causal hop labels. QASPER’s selected subset is yes/no and often one-group, so it tests direct retrieval more than multi-hop closure.

The inherited router is asymmetric. Reusing memory-projected vectors as frontier queries is a clean parameter-free probe, but it may be geometrically mismatched. The experiment does not include evidence-edge training, a learned update, a full agentic baseline, or Level-3 transformer-conditioned traversal. Downstream F1/EM is measured only in a two-example execution smoke and is not used as evidence of quality. Finally, the trace-oriented implementation launches many small operations; production latency requires batched frontiers, batched examples, and a persistent index service.

## 8 Conclusion

Bounded associative closure is a viable retrieval primitive when latent memory contains the right edges. It can find  $B$  through  $A$ , preserve a graph explaining that discovery, and defer full native K/V until search ends. On controlled chains, that behavior is exact. On the current pretrained semantic geometry, however, the result is not yet strong enough to replace one-shot PRA. Hotpot shows a small chain-completion signal alongside semantic drift and substantially more index work; QASPER favors direct retrieval.

The scientific lesson is specific. Iterative machinery alone does not turn nearest-neighbor embeddings into a relational graph. The next step is to learn and falsify associative edges directly, while retaining the same-budget baseline and the strict separation between compact closure and native payload materialization.

## A Implementation Map

The implementation is designed so another system can reproduce the mechanism without adopting the entire PRA-HF package. File and line references below refer to the accompanying source revision.

**Layer-local packed index.** `src/prahf/iterative.py:62--112` packs variable gist sets as `[chunks,max_gists,routing.width]` plus a validity mask. Records retain chunk identities and payload handles, but packing copies no native K/V.

**Graph contract.** `iterative.py:115--170` defines node, edge, graph, and result records. The JSON Schema is `docs/papers/shared/schemas/retrieval_graph_v1.schema.json`. Nodes retain direct score, edge score, path score, hop, parents, winning gist, selection state, and materialization state.

**Exact scoring and deterministic top-k.** `iterative.py:203--224` normalizes frontier rows and computes `einsum("fd,cgd->fcg")` before max reduction across gists. A stable index tie-break is applied before `torch.topk`, producing deterministic identities on CPU and CUDA.

**Traversal.** `iterative.py:226--412` performs root scoring, frontier expansion, anchoring, deduplication, path pruning, Level-2 updates, stopping, and cost accounting. The code never accesses `chunk.token_kv`. Conversion at lines 422–443 returns selected payload handles only after the graph is final.

**Host-model integration.** `src/praxi_hf/model.py:364--411` captures one root query with memory disabled, dispatches one-shot or iterative routing, maps final identities to consumption layers, enables fixed memory, and only then marks graph nodes materialized. Generation thereafter uses the unchanged Paper 2 native-K/V path.

**Reproduction.** The experiment directory is `experiments/paper2_5_iterative_pra/`. Its `run_closure.py` file runs retrieval, `summarize_results.py` derives figures, and `run_downstream_smoke.py` checks CUDA integration. Raw rows, tables, plots, and graph JSONL are in the corresponding `shared/results/paper2_5_iterative_pra/` directory.

## B Validation Gates

The test suite checks  $D = 1$  equivalence to Top- $B$ , deterministic batches, exact synthetic two-hop recovery, deduplication and cycles, depth/beam/unique budgets, all path and frontier modes, zero limits, layer locality, CPU/CUDA identity parity, and the no-materialization-before-closure boundary. The combined Qwen/Llama/Gemma/product/closure subset contains 83 passing tests. The full downstream smoke records zero native operation-limit violations.

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